

Reimagining Robotics Through ARTIFICIAL MUSCLE TECHNOLOGY

Artificial muscles are reshaping robotics, bridging gaps between imagination and reality.
What new frontiers will these innovations unlock?



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Significant progress in robotic muscle technology has enabled the creation of advanced and highly capable robotic systems. These systems, driven by artificial muscles, are engineered to replicate the structure and function of human muscles. However, they go a step further, offering enhanced strength, faster movement, and greater endurance, allowing robots to perform tasks that exceed human physical limitations.

Revolutionising robotics with low-cost actuators

Thanks to the latest advancements in artificial muscle technology, robots can

now move with the grace and flexibility of a worm while maintaining the strength to lift up to 500 grams consistently.

Currently, stiff and rigid actuators used in robotics cost hundreds to thousands of dollars. A flexible, soft actuator costs just \$3 in material (excluding a small motor that alters the actuator's shape to mimic muscle-like effects). This advanced actuator is more practical for a human-centric environment.

The secret lies in 3D-printed soft actuators made from standard rubber. These cylindrical structures, called 'handed shearing auxetics' (HSAs), enable unique movements. Initially, HSAs made from thermoplastic polyurethane—a standard rubber often used in mobile phone cases—were developed. However, these HSAs were challenging to twist into extended and expanded states.

Successful actuation was achieved by assembling two or four HSAs, each equipped with its own motor. This innovative material allows a large muscle capable of withstanding an impressive 5000 lifting cycles without any signs of wear or tear.

The current worm-like robot is compact, measuring just 26 centimetres in length, and can crawl backwards and forward at a speed of over 32 centimetres per minute.

With this breakthrough technology, the possibilities for robotics are endless. From delicate surgical procedures to agile search-and-rescue missions, these artificial muscles are paving the way for a new era of safe and efficient robotics.

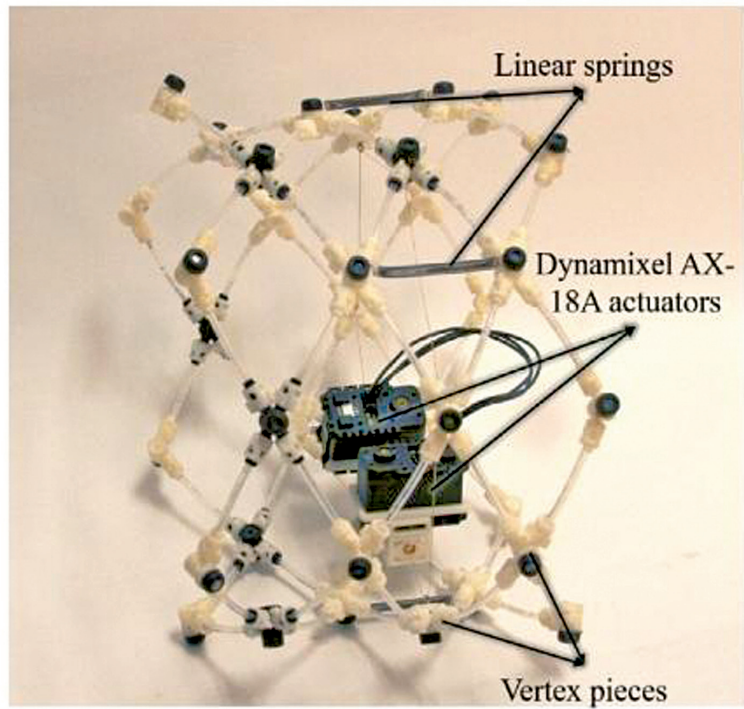


Fig. 1: An actuator that acts as a large muscle for robots (Credit: www.mdpi.com)